🔍 Overview – Computer Forensics

📌 What is Computer Forensics?

Computer Forensics, also known as **Digital Forensics**, is a branch of forensic science involving the **identification, preservation, extraction, documentation, and interpretation of digital evidence** from computer systems and storage devices.

It is used in:

- Cybercrime investigations
- Internal corporate probes
- Civil litigation
- Incident response and breach recovery

The goal is to perform a **structured investigation** while maintaining a documented chain of evidence to uncover what happened on a computing device and who was responsible.

🔹 Difference – Computer Crime & Unauthorized Activities

Aspect	Computer Crime	Unauthorized Activities
Definition	Any illegal act involving computers, networks, or digital devices	Actions violating security policy or acceptable use but not necessarily criminal
Example	Hacking into a bank's system to steal funds	Downloading pirated software in the workplace
Legal Status	Punishable under cyber laws	May or may not be illegal; usually violates company policy
Intent	Usually malicious and planned	Can be due to negligence or lack of awareness
Impact	Legal, financial, and reputational	Operational and disciplinary in nature

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🔹 Cyber Laws

Cyber laws are laws that **govern digital interactions, electronic communication, and internet-based crimes**. They help enforce legal standards related to:

- Intellectual property rights
- Data privacy
- Digital contracts
- Electronic signatures
- Cybercrimes (hacking, phishing, identity theft)

Key Cyber Law Concepts:

1. **IT Act 2000 (India)** – Governs electronic commerce, digital signatures, and cybercrimes.
2.

Data Protection Laws – GDPR (Europe), HIPAA (USA), and PDP Bill (India draft) for privacy.

3. **Cybercrime Laws** – Laws to penalize hacking, unauthorized access, data theft, malware distribution.

Process of Computer Forensics (Six Stages)

1. ## Identification

   - Detecting potential sources of digital evidence.
   - E.g., hard drives, USBs, emails, cloud storage.

2. ## Preservation

   - Ensuring the integrity of digital evidence.

- Imaging the data without altering the original content.

3. ## Collection

   - Acquiring data in a forensically sound manner.
   - Using write blockers and forensic tools (e.g., FTK Imager).

4. ## Examination

   - Filtering relevant data.
   - Recovering deleted files and examining logs.

5. **Analysis**

- Understanding the data and reconstructing events.
- Linking artifacts to suspects or events.

6. **Presentation**

- Reporting the findings clearly and legally.
- Expert witness testimony may be needed in court.

Need for Forensics Investigator

A **Computer Forensics Investigator** is crucial because:

- **Cybercrimes are rising** – Hacking, phishing, ransomware, and insider threats.
- **Digital evidence is fragile** – Can be altered or lost without proper techniques.
- **Legal compliance** – Helps organizations comply with data laws and support investigations.
- **Litigation support** – Provides credible digital evidence in civil and criminal courts.
- **Incident response** – Aids in identifying the root cause and scope of a breach.
- **Internal investigations** – Uncover policy violations or data leaks.

Skills needed:

- Knowledge of operating systems and networks
- Familiarity with forensic tools (e.g., EnCase, Autopsy)
- Understanding of legal procedures